



DELHI PUBLIC SCHOOL NEWTOWN

SESSION: 2021-22

FINAL TERM EXAMINATION

CLASS: IX
SUBJECT: CHEMISTRY

TOTAL MARKS: 80
TIME: 2 HOURS

(Candidates are allowed additional 10 minutes for only reading the papers. They must NOT start writing during this time.)

The Question Paper comprises Two Parts. Part I is based on Multiple Choice Questions.

Part –II is based on Subjective questions.

The intended marks for questions are given in brackets. []

This paper consists of seven printed pages.

Part –I

Attempt all question in Part I

Choose the correct answers to the questions from the given options (a to e). (Do not copy the question, write the correct answer only) [30]

1. A hydrocarbon which is a greenhouse gas is:
 - (a) Acetylene
 - (b) Ethylene
 - (c) Methane
 - (d) Ethane
2. According to Boyle's Law, as the pressure increases, the volume:
 - (a) Increases
 - (b) Decreases
 - (c) Remains same
 - (d) First increases then decreases
3. When a non- metal atom becomes an ion:
 - (a) It loses electron and is oxidized
 - (b) It gains electron and is reduced
 - (c) It loses electron and is reduced
 - (d) It gains electron and is oxidized
4. On adding water to sodium, the solution formed is:
 - (a) Neutral
 - (b) Acidic
 - (c) Amphoteric
 - (d) Alkaline

5. The RMM of ammonium nitrate is 80. The percentage weight of nitrogen in it is:
- (a) 35
 - (b) 17.5
 - (c) 52.5
 - (d) 0.35
6. The metalloid present in period 2 is:
- (a) Silicon
 - (b) Boron
 - (c) Aluminium
 - (d) Carbon
7. A compound has the formula X_2Y , where Y denotes a non-metal. The formula of the compound formed between Calcium and Y will be:
- (a) Ca_2Y
 - (b) CaY_2
 - (c) Y_2Ca
 - (d) CaY
8. The number of electrons and protons present in sulphide ion is:
- (a) 16, 16
 - (b) 18, 16
 - (c) 18, 18
 - (d) 16, 18
9. A metallic nitrate which on decomposition gives a metallic oxide, nitrogen dioxide and oxygen is:
- (a) Sodium nitrate
 - (b) Calcium nitrate
 - (c) Potassium nitrate
 - (d) Silver nitrate
10. When air is blown into a balloon, its pressure ____ and volume ____.
- (a) Increases, decreases
 - (b) Decreases, increases
 - (c) Increases, increases
 - (d) Decreases, decreases
11. The air pollutant which is non-acidic is:
- (a) Nitrogen dioxide
 - (b) Ozone
 - (c) Sulphur dioxide
 - (d) Sulphur trioxide

12. Which one of the following has water of crystallization?
- (a) Potassium chloride
 - (b) Potassium nitrate
 - (c) Washing soda
 - (d) Common salt
13. Drying agent used to dry hydrogen chloride gas is:
- (a) Silica gel
 - (b) Fused calcium oxide
 - (c) Phosphorus pentoxide
 - (d) Concentrated sulphuric acid
14. A sample of a gas has a volume of 500cc at 373K. Calculate the temperature at which the volume becomes 260cc, assuming pressure is constant.
- (a) 94K
 - (b) 79°C
 - (c) -79°C
 - (d) -194°C
15. An atom ${}_{20}\text{Ca}^{40}$ having unstable electronic configuration forms a compound with ${}_8\text{Y}^{16}$. The compound formed is ____ and the formula of the compound is ____.
- (a) Electrovalent, X_2Y
 - (b) Covalent, X_2Y
 - (c) Electrovalent, XY
 - (d) Electrovalent XY_2
16. The formula of sodium borate is:
- (a) $\text{Na}_3(\text{BO})_3$
 - (b) NaBO_3
 - (c) Na_3BO_3
 - (d) NaBO_2
17. Substance used to remove phosphine from hydrogen is:
- (a) Potassium hydroxide
 - (b) Lead nitrate
 - (c) Anhydrous calcium chloride
 - (d) Silver nitrate
18. The reducing agent in the reaction $2\text{HgCl}_2 + \text{SnCl}_2 \rightarrow \text{Hg}_2\text{Cl}_2 + \text{SnCl}_4$ is:
- (a) HgCl_2
 - (b) SnCl_2
 - (c) Hg_2Cl_2
 - (d) SnCl_4

19. The metal above hydrogen in the metal activity series but does not react with dilute acids to liberate hydrogen is:

- (a) Copper
- (b) Lead
- (c) Magnesium
- (d) Zinc

20. Good oxidizing agents are:

- (a) Alkali metals
- (b) Alkaline earth metals
- (c) Halogens
- (d) Inert gases

21. Scandium was originally called:

- (a) Eka-silicon
- (b) Eka-germnum
- (c) Eka- aluminium
- (d) Eka- boron

22. A covalent molecule having two single covalent bonds is:

- (a) Oxygen
- (b) Hydrogen chloride
- (c) Water
- (d) Ammonia

23. A metal which cannot displace copper from copper nitrate solution is:

- (a) Lead
- (b) Silver
- (c) Zinc
- (d) Magnesium

24. The valency of copper in CuS is:

- (a) 1
- (b) 2
- (c) 4
- (d) 0

25. Temporary hardness is due to the presence of calcium and magnesium:

- (a) Chlorides
- (b) Sulphates
- (c) Bicarbonates
- (d) Carbonates

26. Oxygen accepts two electrons and attains the stable electronic configuration of:
- Helium
 - Neon
 - Argon
 - Krypton
27. Carbon and ____ is placed in the same group
- Beryllium
 - Phosphorus
 - Silicon
 - Sulphur
28. The oxidized product formed when hydrogen peroxide is added to potassium iodide is:
- Water
 - Potassium
 - Oxygen
 - Iodine
29. The final volume of a gas, if the pressure of the gas, originally at STP is doubled and its temperature is tripled is:
- Same as original volume
 - Twice of the original volume
 - 0.5 of the original volume
 - 1.5 of the original volume
30. The gas that liberates iodine vapours from potassium iodide solution.
- Nitrogen
 - Nitrogen dioxide
 - Carbon dioxide
 - Sulphur dioxide

Part II

SECTION I (10 Marks)

Attempt all questions from this Section

Question: 1

- a. Name the following: [5]
- The basic radical in K_2MnO_4 .
 - The gas evolved when copper carbonate is heated strongly.
 - A liquid hygroscopic substance.
 - The element present in period 3 and having four valence electrons.
 - The metal which is expensive and not used in the preparation of hydrogen from water.
- b. Complete and balance the following reactions: [5]
- $Al + NaOH + \underline{\hspace{1cm}} \rightarrow \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$
 - $FeSO_4(\text{on heating}) \rightarrow \underline{\hspace{1cm}} + SO_3 + \underline{\hspace{1cm}}$
 - $AgNO_3 (\text{on heating}) \rightarrow \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$
 - $(NH_4)_2Cr_2O_7 \rightarrow \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$
 - $NO_2 + H_2O \rightarrow \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

SECTION II (40 Marks)

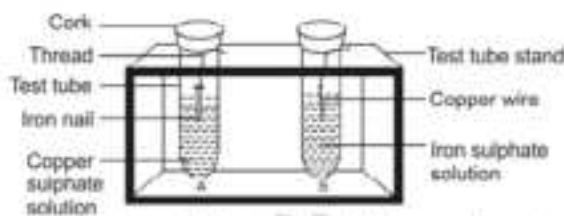
Attempt any four questions from this Section

Question: 2

- a. Hydrogen may be prepared in the laboratory by the action of a metal on an acid.
- Which of the following metals copper, zinc, lead or sodium would be the most suitable one?
 - How would you modify the apparatus to collect hydrogen gas safely?
 - Which drying agent would you employ to collect dry hydrogen gas? [3]
- b. Find the percentage of water in blue vitriol. (Cu=64, S=32, O=16) [3]
- c. Give one example of a photochemical reaction involving- [2]
- Silver salt
 - Water
- d. Complete the following oxidation/ reduction reaction and state which one is oxidation and reduction. [2]
- $\text{Mn}^{7+} \rightarrow \text{Mn}^{5+}$
 - $2\text{Br}^- \rightarrow \text{Br}_2$

Question 3

- a. State if the solubility increases, decrease or remain the same with rise in temperature for each of the following compounds: [3]
- Potassium nitrate
 - Sodium chloride
 - Calcium sulphate
- b. Observe the two test tubes A and B in the diagram given below and answer the following questions: [3]



- In which test tube will the reaction take place?
 - Write a balanced equation of the reaction that will take place.
 - Name the type of reaction.
- c. Write a balance equation for the removal of: [2]
- Permanent hardness
 - Temporary hardness by addition of washing soda (in both cases)
- d. Name a covalent molecule having two lone pair of electrons. Represent it with an orbit diagram. [2]

Question 4

- a. At a constant temperature, a gas at a pressure of 750mm pressure occupies a volume of 100cm^3 . If the volume is decreased by 40%, then find the new pressure. [3]
- b. Write the equations involved in the production of sulphur dioxide in the atmosphere that causes acid rain. [3]

- c. Give reasons for the following: [2]
- Dobereiner's method of classification of elements did not hold much weightage for future classification.
 - Sodium is univalent.
- d. Give the chemical name of: [2]
- CaC_2O_4 .
 - NaClO

Question 5

- a. State what do you observe: [3]
- Iodine crystals are heated in a test tube
 - Ferric chloride crystals are exposed to atmosphere for some time.
 - Sodium hydroxide solution is added to ferrous sulphate solution dropwise, and then in excess.
- b. A metal in period 3 burns with a lilac flame in air. It combines with oxygen and form a strong alkali. [3]
- Name the metal.
 - What is its atomic number?
 - Write the formula of its plumbate.
- c. 6dm^3 of dry gas is collected at a temperature of 27°C and pressure 700mm Hg. Find the volume of the gas at STP. [2]
- d. Give reasons for the following: [2]
- Acid rain causes nutrient leaching.
 - Magnesium reacts with very dilute nitric acid at low temperatures liberating hydrogen.

Question: 6

- a. Name the element present in: [3]
- Group 1 and period 2
 - Group 2 and period 3
 - Group 14 and period 2
- b. Atomic numbers of the elements A to E are given as: A= 7, B= 8, C=9, D=12, E=19
- Which one is a divalent metal?
 - What type of compound will it (metal in i) form with C?
 - Draw an electron dot diagram to represent the compound formed in (ii) above.[3]
- c. On heating a blue coloured powder of copper salt in a boiling tube, a black substance is left behind with the evolution of a colourless gas that rekindles a glowing splinter and on heating strongly a reddish brown gas is evolved. [2]
- Write a balanced chemical equation for the above reaction.
 - Identify the type of reaction.
- d. Write the molecular formula for the following compounds: [2]
- Lead chromate
 - Aluminium carbide.